Wonguk Lee

Ph.D. Student, Management Information Systems (MIS)
Business and Technology Management (BTM), KAIST College of Business
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1. EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)**

  Ph.D. Student in Business and Technology Management (BTM)
  Major: Management Information Systems (MIS)

- **Korea Advanced Institute of Science and Technology (KAIST)**

  M.S. in Business and Technology Management (BTM)
  Major: Management Information Systems (MIS)

- **Handong Global University (HGU)**

  B.B.A. in Business Administration
  B.A. in Law (Double Major)

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2. RESEARCH INTERESTS

- GIS, Mobility and Geographic Access to Healthcare
- Technology Adoption, Human–AI Interaction and Social Outcomes
- Conversational AI, Mental Health and Perceived Judgment
- Data Ontologies, Multi-Agent Systems and Generative AI

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3. INDUSTRY EXPERIENCE

- **Chief Data Officer (CDO), TalentSeeker**

  Feb 2024 – Present | Full-time C-level role
  Lead data ontology development, multi-agent systems, data-pipeline architecture, data engineering, and LLMOps; translate research into AI-enabled recruitment systems.

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4. TEACHING EXPERIENCE

- **Lead Course Instructor in Data Science, Baekseok University**

  2026 academic year, second semester | Big Data / Data Science | English-medium instruction
  Responsible for course design and delivery covering Python, pandas, data preparation, visualization, basic statistics and prediction, generative AI verification, and data storytelling.

5. PROJECTS

- **Smart Mobility Safety Services and Infrastructure for Autonomous Driving**

  MOTIE / KIAT, International Industrial Technology Cooperation | Project P0025238
  2025 | Technical development and patent contribution
  Developed OSM-supported road-detection and road-geometry algorithms and physics/XGBoost-based speed recommendations with KAIST and industry collaborators.

- **Public Bike-Sharing and Transit Integration for Local Mobility and Consumer Activity**

  Yuseong District Office and Daejeon Transportation Corporation support | Project lead
  Secured and led the project; integrated Tashu bike-sharing, transit, Shinhan Card spending, and KT population-movement data to study first/last-mile connections, local festivals, and economic activity.

- **Personal Mobility Device Adoption among Users and Non-users**

  Behavioral research | Manuscript under review
  PLS-SEM and multi-group analysis of 444 respondents, including 222 users and 222 non-users; examined perceived physical risk, trust, and adoption intentions.

- **AI Counseling Chatbot for Schizophrenia Care**

  Jeonbuk National University Hospital | Ongoing
  Develop and study a free patient-facing LLM/RAG chatbot, focusing on perceived judgment, stigma-related engagement barriers, disclosure, and supportive interaction design.

- **Onto-LLM-Based Global Outbound Recruitment Service Development**

  TIPS R&D Program, Ministry of SMEs and Startups (MSS) | Aug 2025 – Jul 2027 | Ongoing
  Contribute to ontology-based recruitment AI, Graph-RAG / Graph-of-Thought reasoning, and agent prototype research.

- **Open-Web-Based Automated Resume Collection and AI Recruitment Support System**

  Startup Growth Technology Development Program, TIPA / MSS | Aug 2024 – Jul 2025 | Completed
  AI recruitment and data-pipeline development.

- **Multiple-RAG and Multidimensional Ontology-Based Recruitment sLLM**

  Didimdol R&D planning | 2026
  Research and engineering design for a recruitment-specific small language model, evidence retrieval, ontology mapping, evaluation, and an end-to-end processing pipeline.

- **SF-Olive Smart Farm Platform for Olive Production**

  International Collaborative Technology Development | Planned project period Aug 2026 – Jul 2029
  KAIST participating researcher role: data experimentation and bias-analysis support for a platform combining advanced sensing, blockchain, and edge AI semiconductors.

6. PUBLICATIONS AND WRITING

- **Book Chapter**

Lee, W. (2026).
"Generative AI as a Creative Collaborator."
In *AI Transformation Era: Innovation and Challenges of Korean Industries Highlighted by KAIST*.
KAIST Graduate School of Technology and Management, Yulgok Publishing.

- **Academic Book (Co-author)**

AI Era: Structure and Sustainability of the Korean Workforce Ecosystem
Selected as an Excellent Academic Book by the National Academy of Sciences, Republic of Korea (2025)

- **Research Papers and Manuscripts**

(1) Understanding Personal Mobility Device Adoption Through the S–O–R Framework: A Multi-Group Analysis of Users and Non-users

Under review. Uses PLS-SEM and multi-group analysis to examine trust, physical risk, and adoption intentions among users and non-users.

(2) The Effects of Public Bike-Sharing (Tashu) and Public Transit Integration on Local Festivals and Consumer Spending

Under revision at a Korean journal. Excellent Paper Award, Korea Society of IT Services. Examines transit complementarity, local movement, and consumer activity.

(3) Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts

Examines how thought-bubble transparency in AI counseling systems influences perceived judgmentalness, trust, disclosure intention, and compliance intention in mental health contexts.

(4) The Role of NFTs in the Digital World: Catalyzing Innovation in the Creative Industry through a Patent-Inspired Approach

Explores the role of NFTs in digital innovation and creative industry transformation through a patent-inspired analytical framework.

(5) When the AI Bot Advisor Leaves: Unpacking Investor Shifts and Loan Imbalances in P2P Lending

Investigates how the withdrawal of AI bot advisors affects investor behavior and loan allocation dynamics in P2P lending platforms.

- **Conference Research Dissemination**

CIST; International Conference on Information Systems (ICIS); Symposium on Statistical Challenges in Electronic Commerce Research (SCECR).
Research on conversational AI, transparency, and mental-health interaction.

7. PATENTS

- **Road Determination Device and Method of Determining a Road**

  Korean Patent Application No. 10-2025-0209757
  Filed: Dec 24, 2025

- **Recommended Speed Determination Apparatus and Method**

  Korean Patent Application No. 10-2025-0207816
  Filed: Dec 23, 2025
  Related technical work: physics- and XGBoost-based recommended-speed calculation for road segments.

- **Ontology-Based Data Standardization Apparatus for Human Resource Management and Operating Method**

  Korean Patent No. 10-2891011
  Filed: Jul 24, 2025 / Registered: Nov 20, 2025

- **Apparatus and Method for Ontology-Based Candidate Search**

  Korean Patent Application No. 10-2025-0170399
  Filed: Nov 12, 2025

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8. HONORS AND AWARDS

- **Excellent Paper Award**

  Korea Society of IT Services | Research on public bike-sharing, public transit integration, local festivals, and consumer spending

- **Outstanding Research Idea Award**

  BTM Seminar, KAIST | When the AI Bot Advisor Leaves: Unpacking Investor Shifts and Loan Imbalances in P2P Lending

- **Excellent Academic Book Selection (National Academy of Sciences, Republic of Korea)**

  AI Era: Structure and Sustainability of the Korean Workforce Ecosystem (Co-author)

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9. RESEARCH AND TECHNICAL METHODS

- GIS and OSM road networks; satellite-image analysis; road segmentation; physics-based modeling; XGBoost
- PLS-SEM; multi-group analysis; survey and experimental research; mobility and spending-data integration
- Python; pandas; data engineering; ontologies; RAG; multi-agent systems; LLMOps